

Lesson Plan: Exploring Electric Charge Through Tape & Balloons

DLE 915 Core Assignment #3 Part 1: A-B-C-D Lesson Plan Checklist

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Focal EL Level: Bridging

A. Relationship Between Content Standards and EL Standards

(A1) Content Standards and Lesson Objectives

Content Standard: HS-PS3-5: Develop and use a model of two objects interacting through electric or magnetic fields to illustrate the forces between objects and the changes in energy of the objects due to the interaction.

Supporting Standard: HS-PS2-4: Use mathematical representations of Newton's Law of Gravitation and Coulomb's Law to describe and predict the gravitational and electrostatic forces between objects (qualitative/conceptual in this lesson).

Content Objectives:

- Observe and record the behavior of tape strips and balloons charged through different methods.
- Identify patterns in attractive and repulsive interactions between charged objects and between charged and neutral objects.
- Construct an initial model (drawing) that represents their thinking about what causes charge interactions.
- Predict the behavior of charged objects in novel situations based on observed patterns.

(A2) ELD Standards Alignment - Bridging Level

ELD Part I, Standard 1 (Exchanging Information and Ideas) - Bridging: Students contribute substantively to academic conversations by building on the ideas of others, expressing their own ideas clearly with precision, and negotiating meaning. *[This standard connects to the lesson's content objectives because students must describe and explain their observations of charge interactions during partner talk and whole-class share, using evidence-based academic language.]*

ELD Part I, Standard 10a (Writing) - Bridging: Students write longer and more detailed literary and informational texts collaboratively and independently using appropriate text organization and register. *[This standard connects to the content objective of constructing an initial model, as students must label their visual model with target vocabulary and produce written explanations in their response documents.]*

ELD Part II, Standard 1 (Understanding Text Structure) - Bridging: Students apply understanding of how text types are organized to comprehending and composing texts with increasing independence. *[This standard connects to the discourse structure of the lesson, where students move from observation notes to explanatory writing to visual models—the structure of scientific explanation.]*

B. Relationship Between Content Objective/Standard and Instruction

(B1) How Instructional Activities Support the Content Objective

The lesson follows an **Explore–Summarize** structure designed to build conceptual understanding from direct experience. Students first observe real phenomena (tape strips repelling or attracting, balloons repelling each other, a charged balloon rolling an aluminum can) and record their observations in a digital response document. Each investigation trial follows a consistent routine: students attempt the activity themselves (~45 sec), discuss with a partner, watch the teacher demonstrate with narration, and then record observations and explanations. This cycle ensures that every student, regardless of language level, has firsthand sensory experience with the phenomenon before being asked to describe or explain it. In the Summarize phase, students construct an initial model (a drawing of the invisible mechanism causing attraction or repulsion), which directly serves the content standard’s emphasis on developing and using models. The activities are not merely engaging, each one builds toward the objective of identifying patterns in charge interactions and constructing an explanatory model.

(B2) Modalities and Language Load for the Bridging Level

The lesson incorporates multiple modalities to differentiate for the Bridging-level EL student: **visual** (observing demos, viewing sentence frames on the board), **kinesthetic** (handling tape and balloons), **verbal** (partner talk and whole-class share), **written** (recording observations in the

response document), and **graphic** (model sketching). For a Bridging student, the language load is moderate to high: they are expected to use academic vocabulary and produce written explanations, but the sentence frames reduce syntactic demand, and the visual model sketch provides an alternative mode of expression that lowers the language barrier while maintaining cognitive rigor. The text type is a **structured response document** (fill-in with sentence starters), which is appropriate for Bridging students because it provides enough scaffolding to guide academic writing without over-simplifying the task.

(B3) Modifications to Encourage EL Participation

The lesson uses **assigned partner pairs** for all investigation trials, with intentional pairing so the Bridging student is placed with a supportive peer who models academic language. Group size is small (pairs), which maximizes talk time and reduces the anxiety of speaking in a large group. The **independent exploration** → **partner talk** → **whole-class share** structure gives the Bridging student processing time before public speaking. The teacher uses **cold-calling strategically** during whole-class share, giving the Bridging student the opportunity to contribute after they have already rehearsed their response with a partner. The time frame for each activity is structured and brief (~30 seconds for hands-on, ~2 minutes for partner talk), which keeps the pacing manageable.

(B4) Modified Materials

Materials are modified in several ways for the Bridging-level EL: the **digital response document** includes built-in sentence starters that scaffold written responses. **Sentence frames are posted on the board** throughout the lesson for ongoing reference (e.g., “When I bring ____ near ____, I observe that ____”). The **model sketch** allows students to express conceptual understanding visually, with labels, rather than relying solely on written paragraphs. All vocabulary is introduced with **gesture and demonstration** alongside verbal definition.

(B5) Strategy from 50 Strategies Textbook

This lesson incorporates Strategy 10: Skills Grouping from Herrell and Jordan's *50 Strategies for Teaching English Language Learners* (2003). Skills grouping involves intentionally organizing students into small groups based on their current language or content needs, allowing the teacher to target instruction more precisely. In this lesson, the use of assigned partner pairs during

investigation trials reflects this strategy: the Bridging-level EL students are intentionally paired with a peer who models academic language consistently, creating a small collaborative unit where the focal student hears and practices precise scientific vocabulary in context. Rather than grouping students randomly or by friendship, the teacher must make deliberate pairing decisions based on language proficiency and content understanding. This is particularly appropriate for a Bridging student who already has strong conversational skills. The strategic pairing pushes them beyond social language and into the academic register by placing them alongside a peer who naturally uses terms like "attract," "repel," and "transfer" rather than everyday equivalents.

C. Relationship Between Language Demands

(C1) Language Function

The primary language function in this lesson is **describe**. Students describe observed charge interactions using evidence-based language throughout the Explore and Summarize phases. A secondary language function is **predict**—students predict the behavior of charged objects in novel situations and justify their predictions using observed patterns. For the Bridging-level student, these functions are scaffolded through the sentence frames, which model the grammatical structures needed to perform each function: “When I bring ____ near ____, I observe that ____” (describe) and “I predict that ____ will ____ because ____” (predict).

(C2) Vocabulary

Tier 1 (Basic): pull, push, stick, move, touch, near, far

Tier 2 (General Academic): observe, predict, model, evidence, pattern, interact, demonstrate, compare

Tier 3 (Content-Specific): charge, positive (+), negative (−), attract, repel, neutral, transfer, charging, electric field

Bridging-Level Vocabulary Modification: The Bridging student has conversational access to many Tier 1 words but needs support transitioning from everyday equivalents (“stick together,” “push away”) to precise Tier 3 terms (“attract,” “repel”). Vocabulary is introduced through gesture and demonstration during the teacher demo, reinforced through the sentence frames (which embed Tier 3 terms in context), and available on the board throughout the lesson. The

teacher explicitly connects Tier 1 to Tier 3 during instruction (e.g., “When we say they ‘stick together,’ the scientific word is ‘attract’”).

(C3) Syntax

The lesson addresses syntax through **sentence frames** that model the complex sentence structures required for scientific explanation at the Bridging level:

- “*When I bring ____ near ____, I observe that ____.*” (temporal clause + observation)
- “*I predict that ____ will ____ because ____.*” (prediction + causal reasoning)
- “*The two objects ____ (attract/repel) each other, which suggests ____.*” (observation + inference)

These frames are designed specifically for a Bridging student who can produce conversational sentences fluently but needs scaffolding for the subordinating conjunctions and causal connectors that academic scientific writing demands. The frames are displayed on the board as a reference, not a requirement, allowing the Bridging student to use them as needed while moving toward independent production.

(C4) Discourse

Discourse in this lesson is organized through the **independent exploration → partner talk → whole-class share** structure, which mirrors how scientific discourse is organized in the field: individual observation/thinking, followed by peer discussion, followed by community sharing and consensus-building. Written responses in the digital document move from observation to explanation, and the model sketch provides visual discourse—a non-linguistic mode of organizing and communicating scientific ideas. For the Bridging student, this discourse structure is particularly effective because it provides multiple low-stakes opportunities to practice academic language before performing, and it mirrors the structure of scientific argumentation that the student will encounter in more formal writing tasks later in the unit.

D. Relationship Between Content Standard and Assessment

(D1) Before the Lesson - Prior Knowledge Assessment

Assessment: Opening Connect prompt - “Recall a time you felt or experienced static electricity. What did it feel like? What did it look like?” Students write silently, then share with a partner and whole class.

Assumed prior knowledge: Everyday experience with static electricity (shocks, hair standing up, clothes sticking). No formal physics content knowledge is assumed.

Assessing Prior Knowledge: Teacher listens, during turn-and-talk and whole-class share, for existing conceptions and vocabulary. This is an informal check to gauge what language and ideas students bring to the topic.

EL Modification: The prompt draws on personal, sensory experience rather than academic knowledge, which is accessible to the Bridging student. The independent exploration provides processing time, and the partner talk allows the student to rehearse their response before sharing publicly.

(D2) During the Lesson - Formative Assessment

Assessment: Teacher circulates during partner talk and investigation trials, listening for target vocabulary use and evidence-based explanations. Teacher tracks whole-class share responses to inform Lesson 2 planning.

Content objective assessed: Identifying patterns in attractive and repulsive interactions; using target vocabulary (charge, attract, repel, neutral) to describe observations.

EL Modification: For the Bridging student, the teacher specifically listens for whether the student is transitioning from everyday language (“they stick”) to academic language (“they attract”). The teacher provides real-time verbal feedback during circulation, affirming accurate vocabulary use and prompting deeper thinking (“You noticed they attract, what might cause that?”). The sentence frames on the board serve as an in-the-moment scaffold the student can reference during partner talk.

(D3) After the Lesson - Individual Assessment

Assessment: Individual model sketch, collected at the end of the lesson. Evaluation criteria: (1) observations are specific and use target vocabulary; (2) explanations connect observations to an underlying cause; (3) model sketch represents an invisible mechanism.

EL Modification: The model sketch is a visual assessment that allows the Bridging student to demonstrate conceptual understanding through drawing and labeling rather than relying solely on extended written prose. The student is expected to use target vocabulary in their labels and any written annotations, but the primary mode of expression is visual, which reduces the language barrier while maintaining the cognitive demand of the content standard (developing and using a model). The teacher provides brief written feedback on the sketch that notes observation strengths and poses a guiding question for Lesson 2, giving the Bridging student specific, actionable language input.